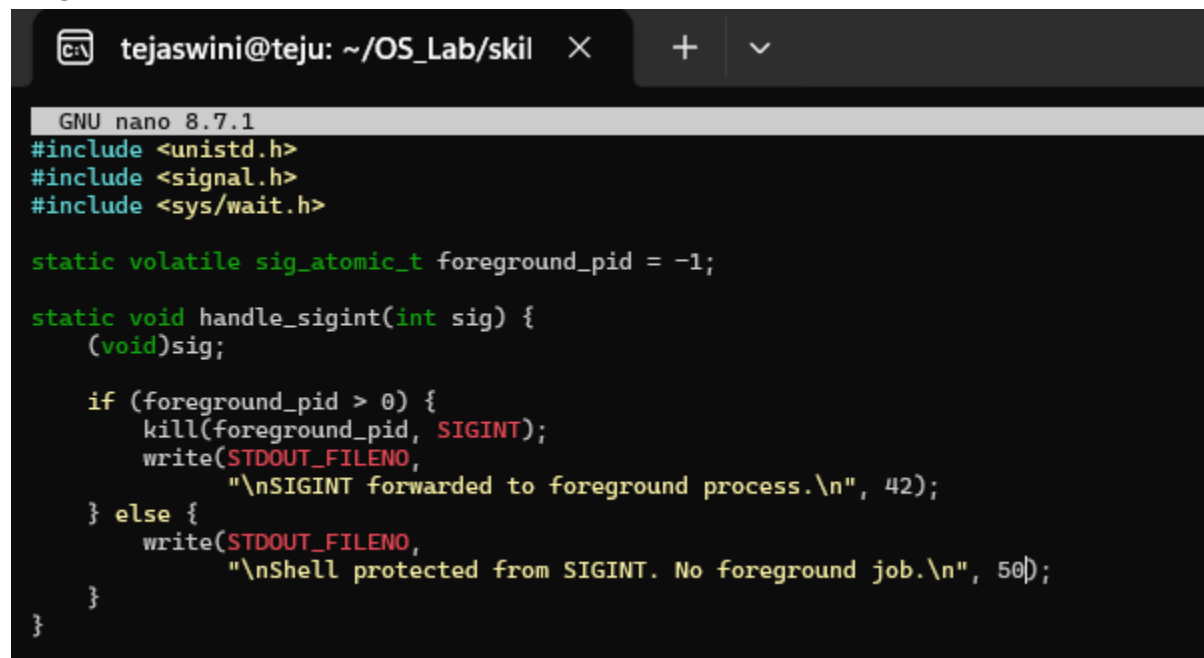


Skill-19

Rollno:2520030011

Program:



```
tejaswini@teju: ~/OS_Lab/skil × + ∨
GNU nano 8.7.1
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

static volatile sig_atomic_t foreground_pid = -1;

static void handle_sigint(int sig) {
    (void)sig;

    if (foreground_pid > 0) {
        kill(foreground_pid, SIGINT);
        write(STDOUT_FILENO,
            "\nSIGINT forwarded to foreground process.\n", 42);
    } else {
        write(STDOUT_FILENO,
            "\nShell protected from SIGINT. No foreground job.\n", 50);
    }
}
```

```

int main(void) {
    struct sigaction sa;
    sa.sa_handler = handle_sigint;
    sigemptyset(&sa.sa_mask);
    sa.sa_flags = 0;

    if (sigaction(SIGINT, &sa, NULL) == -1) {
        perror("sigaction");
        return 1;
    }

    printf("=== Skill 19: SIGINT Handler ===\n");
    printf("The parent handles Ctrl+C and forwards it to the child.\n");

    pid_t pid = fork();

    if (pid == -1) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        signal(SIGINT, SIG_DFL);
        printf("Foreground child pid=%d. Press Ctrl+C.\n", getpid());
        fflush(stdout);

        for (;;) {
            printf("Child running...\n");
            fflush(stdout);
            sleep(1);
        }
    }

    foreground_pid = pid;

    int status;
    waitpid(pid, &status, 0);

    foreground_pid = -1;

    printf("Foreground child terminated. Shell continues safely.\n");
    return 0;
}

```

Output:

```

tejaswini@teju:~/OS_Lab/skill$ nano skill_19.c.c
tejaswini@teju:~/OS_Lab/skill$ gcc skill_19.c.c -o skill_19
tejaswini@teju:~/OS_Lab/skill$ ./skill_19
=== Skill 19: SIGINT Handler ===
The parent handles Ctrl+C and forwards it to the child.
Foreground child pid=1451. Press Ctrl+C.
Child running...
Child running...
Child running...
Child running...
Child running...
Child running...
^C
SIGINT forwarded to foreground process.
Foreground child terminated. Shell continues safely.
tejaswini@teju:~/OS_Lab/skill$ |

```